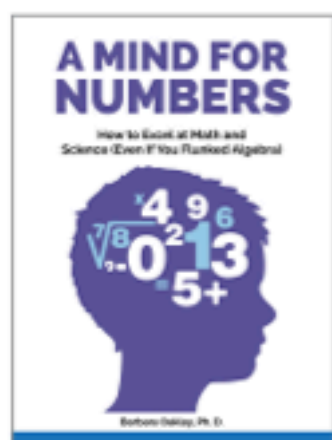




A MIND FOR NUMBERS

How to Excel at Math and Science
(Even If You Flunked Algebra)

BARBARA OAKLEY, PH.D



The Big “So What”

This book explains powerful learning concepts and techniques to help you improve in math, science, and virtually any topic/area in life. The insights are relevant for students and learners of all ages who wish to learn better, faster and deeper.



Introduction



As a child, Oakley disliked math and science because she couldn't grasp the technical details. After she started work, she realized that those inadequacies were limiting her career opportunities. So, she decided to rewire her brain and teach herself to get good at math and science. The more progress she made, the more she started to enjoy the subjects and the better/faster she learned.



Oakley is now a professor of engineering, and actively researches different areas ranging from engineering to humanities and social sciences. In this book, she shares learning insights from scientists, educators, students, famous inventors (such as Thomas Edison and Richard Feynman), as well as her personal experiences. In this summary, we've organized the key ideas into 2 parts:

- How the brain works; and
- Strategies, concepts and ingredients for learning.

KEY QUOTES

"If you've ever wanted to become better at anything, this book will help serve as your guide."

"My success reinforced my desire to practice, and that built more success."

"What you discover will help you be more effective and creative, not only in math and science, but in almost everything you do."



Understanding How Your Brain Works

Your brain is capable of handling complex mental calculations and problems. Each day, you're already making countless unconscious decisions involving distances, speed, and quantities. **To get better at math and science, you must learn how your brain works so you can use it to learn faster and deeper.**

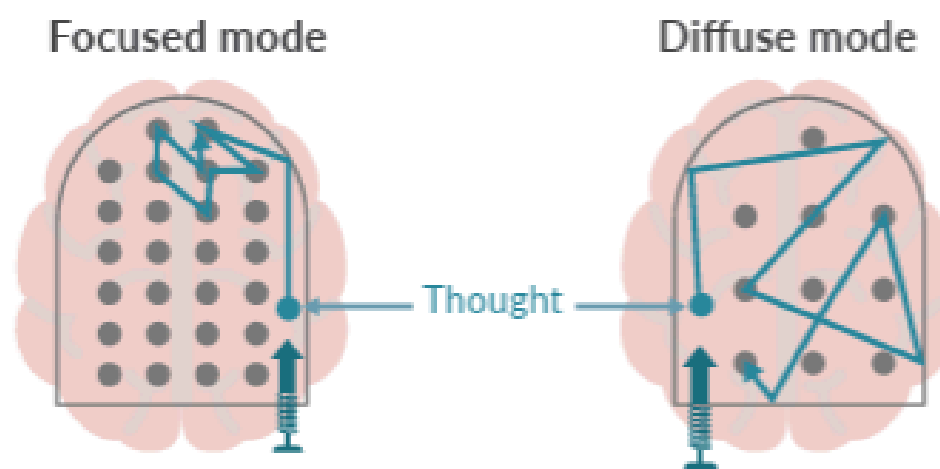


Focused vs Diffused Thinking



Neuroscientists found that our brain switches frequently between 2 sets of thought processes: focused thinking and diffused thinking.

- The focused mode concentrates your prefrontal cortex (which handles logical, analytical thinking) to consciously target a specific issue. It's like shining a strong beam of flashlight in the dark, and is great for deep work or execution.
- The diffuse mode occurs in a more relaxed state. Your mind starts to wander, picking up scattered thoughts and ideas from different parts of the brain, and combining them creatively to arrive at fresh insights. This often happens in the background without your conscious awareness.
- When your brain releases a thought, it's like a ball being released into a pinball game.



- (i) The *focused mode* is great for things that you're familiar with. The concepts are already tightly-connected in your mind—like a pinball game with bumpers close together—so the ball bounces rapidly between the bumpers.

KEY QUOTES

"We all have a natural feel and flair for math and science. Basically, we just need to master the lingo and culture."

"Most difficult problems are solved through intuition, because they make a leap away from what you are familiar with."

- (ii) The *diffuse mode* is better for big-picture thinking when you're exploring new or unfamiliar concepts. The bumpers are further apart so the ball travels a longer distance, allowing you to explore new ideas or approaches.



In short, the diffuse mode allows you to quickly scan for new inputs, while the focused mode allows you to get things done. We need both modes of thinking to learn and survive.

- However, you can't apply both modes concurrently. You can only toggle repeatedly between the 2 modes: scan for data → process/integrate it → scan for more data/ideas.
- To learn math and science, you need the focused mode to analyze a problem rationally and sequentially, and the diffuse mode to grasp abstract concepts like integrals or electric currents.



The 2 Key Memory Systems



Learning involves 2 key memory systems: your short-term and long-term memory. These are like your computer's random access memory (RAM) and hard-drive.

- Your short-term memory or working memory holds whatever information you're processing consciously. It has a limited capacity, and can hold about 4 main clusters of information at any point in time, like a juggler who can keep only 4 balls in the air.
- Your long-term memory is like a warehouse that can store huge amounts of material. However, it can be hard to find a parcel of information in a room filled with parcels. To keep the memories accessible, you need to revisit them occasionally.



To learn effectively, improve your memory and how you transfer information from the short-term to long-term memory.



Learning Strategies, Concepts and Ingredients

In this section, we'll explain various interconnected concepts about learning, and how they can be applied to math and science.

KEY QUOTES

"To learn about and be creative in math and science, we need to strengthen and use both the focused and diffuse modes."



Distributed Practice



Math and science are best learned in small daily doses so your brain has the time to absorb and internalize what you've learned. When you're learning something new, it's crucial to alternate between focused and diffused thinking. This is also known as "distributed practice", i.e. taking breaks between learning sessions instead of doing mass-learning.

- Start with the focused mode. Study intensely about the subject, then relax or do something totally different, e.g. take a nap, listen to music, exercise, play video games, or check your emails.
 - (i) In the diffuse mode, you seem relaxed but your brain is still working in the background to synthesize ideas and generate fresh insights. That's why creative ideas often pop up after a break.
 - (ii) However, you can't create something out of nothing. Start with focused learning to provide "raw materials" for your brain to work with.
- Pause just long enough for your brain to stop thinking about the problem consciously, but not so long that you forget what you've learned. If you're learning something new, always return to the materials within a day.
- Famous inventor Thomas Edison was known for taking regular naps. When he was stuck with a tough problem, he'd take a nap with a ball bearing in his hands. As he fell asleep (which meant that his brain had relaxed to a diffuse mode), the ball bearing would clatter to the floor and wake him up. Edison would then gather any new thoughts/ideas and continue his work.



Moreover, you need mental energy to focus. Without periodic breaks, you'll deplete your mental resources quickly.

- It's similar to physical training. If you lift weights non-stop, you'll exhaust or even injure your muscles. The best way to build muscles is to exercise intensely, then give your muscles time to recover and grow.

KEY QUOTES

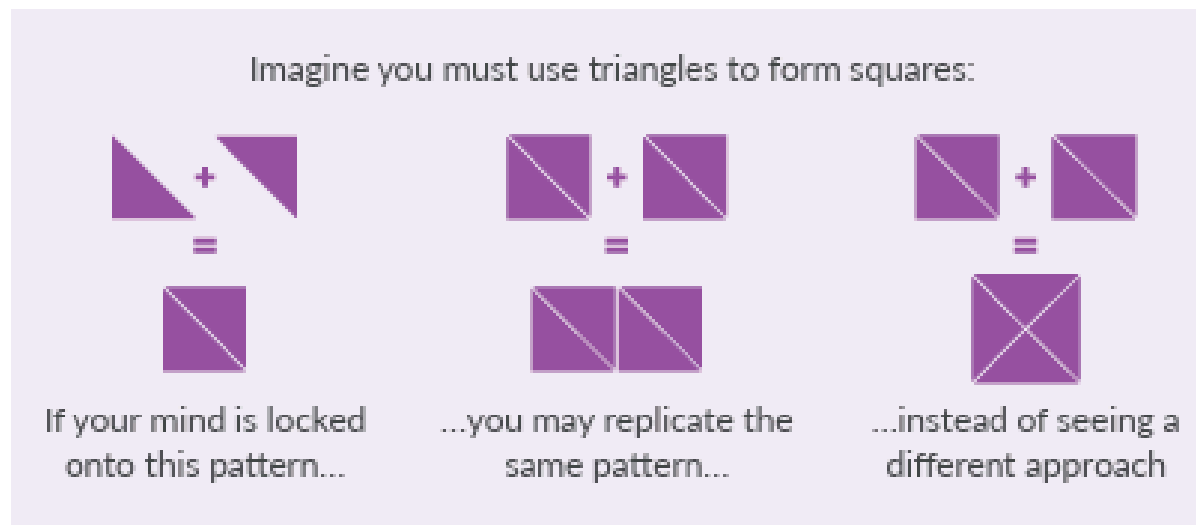
"It's best to work at math and science in small doses—a little every day...That's how solid neural structures are built."

"Sometimes...your desire to figure things out right now is what prevents you from being able to figure things out."

- Feelings of frustration are a signal to shift to diffuse mode.



The Einstellung effect refers to our tendency to get stuck because we get fixated on a wrong approach or a flawed concept, thus blocking out better approaches or solutions.



- When dealing with new or complex problems, don't latch on to a solution too quickly. Study the problem in detail, then step back to see the big picture. You can blink or close your eyes to momentarily disconnect from the problem in front of you.
- Check your work and assumptions. Rushing ahead with a wrong calculation or assumption will only slow you down.
- Brainstorm with others to uncover blind spots, erroneous thinking or assumptions. Be prepared to unlearn old concepts in order to learn new ones.



Chunking and Practice



To get good at math and science, you must grasp the concepts, equations/formulas, then recall and apply them to different types of problems. This requires both memory and understanding.



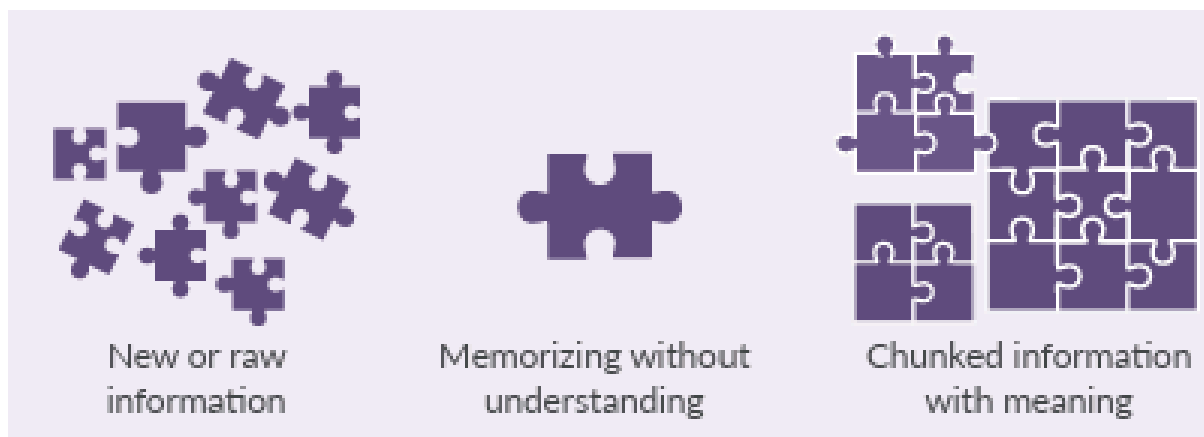
Chunks are bits of information that are connected by meaning. For example, the word "sum" comes from the letters "s", "u" and "m". To do a "golf swing" you must string together a series of steps from positioning yourself to moving your arms, torso, and hips.

KEY QUOTES

"Focused attention...can often help solve problems, but it can also create problems by blocking our ability to see new solutions."

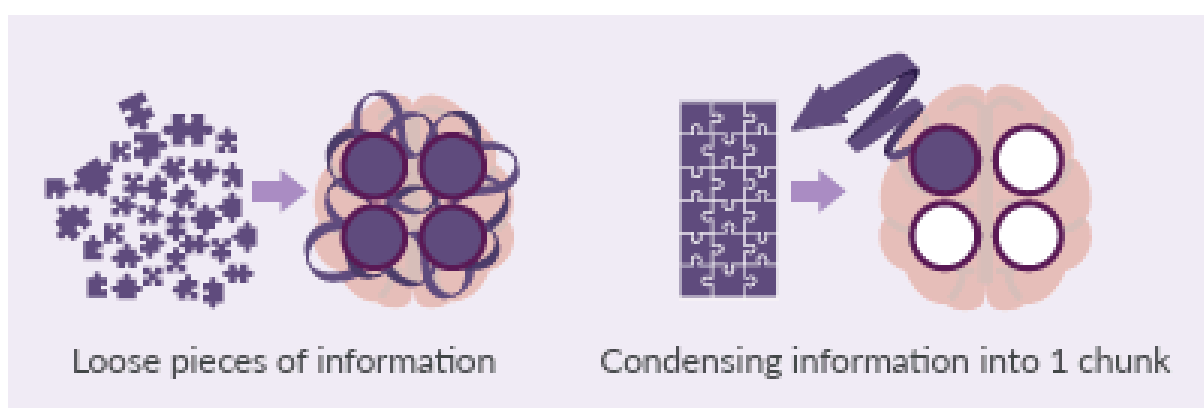
"You have to unlearn your erroneous older ideas even while you're learning new ones."

- When you're learning something new, you must dissect each component, which makes it effortful. For instance, to play the piano, you must learn the notes of the keyboard, the correct fingering, and how to read the music score.
- With enough practice, your brain forms neural patterns or conceptual chunks so you no longer need to remember every detail. When you think of "playing the piano", your brain pulls out a chunk of information instead of separate components.



Chunking frees up your available working memory.

- Initially, each loose piece of information takes up a part of your working memory. This maxes out your short-term memory, which can only hold about 4 clusters of information.
- When you condense the information into 1 chunk, it (i) frees up the other 3 clusters of working memory, and (ii) creates a string of ideas so it's easier to link to other ideas.



The more you practice chunking, more easily and intuitively you can solve problems because (i) you have a larger library of mental chunks, (ii) each chunk contains more data (or a longer "string of data"), and (iii) it's easier to connect between chunks. Each new math/science concept builds your mental scaffold until you can intuitively solve problems at a glance.



Generally, chunks are built in 3 steps:

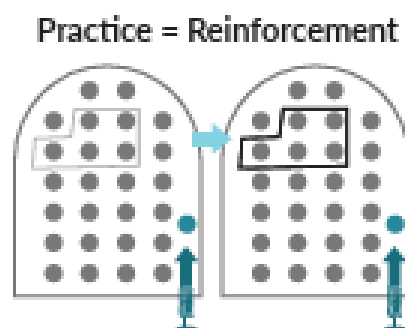
KEY QUOTES

"Chunking...is the mental leap that helps you unite bits of information together through meaning."

"Practice helps build strong neural patterns—that is, conceptual chunks of understanding."

- Remove all distractions to focus fully on the material.
- Understand the basics, e.g. the concept of atoms or how to solve a specific type of math problem.

- Keep practicing to reinforce the neural pathways. Apply the insights across different situations until you gain context and see the big picture. At this point, you will know:



- How to use a method in different scenarios (*bottom-up chunking*); and
- Where a chunk fits in the big picture, i.e. when a method will or won't work (*top-down chunking*).



Initially, you must practice at least daily to reinforce a new pattern/chunk before it fades away. Once you've grasped the concept, you can refresh the idea every few weeks, then every few months or years.

- The bigger your library of chunked concepts and solutions, the more easily you can solve new problems.
- Repeatedly using an incorrect method can build wrong chunks, so make sure you check your work to get it right from the start.
- If you're stuck, ask someone for their inputs or do some generic research (e.g. who invented that method and why). The new inputs and perspectives can help you to see things differently and develop your chunks.



The more effort you put into learning the material, the more deeply it gets embedded in your memory. To accelerate chunking, use retrieval practice (i.e. recall or mental retrieval).

- Don't just passively reread your material. Instead:
 - After each page or chapter, look away and *recall* the key points; or
 - Used *spaced repetition*, i.e. wait for some time before you reread the material. As you walk into your classroom, try to recall the key ideas from the last lesson.

KEY QUOTES

"In some sense, recall helps build neural hooks that you can hang your thinking on."

"Context is where bottom-up and top-down learning meet."

"A key to building mental flexibility and expertise is to build your library of chunked solution patterns. This is your rapid-access data bank—always handy in a pinch."

"The more you do it, the easier it becomes, and the more useful it is."

- Be selective in what you highlight or mark in your texts. Highlight/mark no more than 1 sentence per paragraph, and ideally summarize the ideas in your own words in the margin.
- Rewrite your lecture notes each evening to reinforce the new chunks being formed, or to uncover gaps to be filled. Writing by hand facilitates learning more than typing.
- Try to solve the problems on your own before checking the answers. The more deeply you think about the problem, the more you strengthen the associated neural networks.



When you are building a chunked library, it helps to tackle the toughest aspects using deliberate practice.

- First, work on 1 problem repeatedly:
 - (i) Work on the problem on paper without referring to notes or answers.
 - (ii) Rework the problem, paying attention to your processes.
 - (iii) Take a break or do something totally different.
 - (iv) Rework the problem just before bed. Listen to your intuition.
 - (v) The next day, rework the problem again. By now, it should seem relatively effortless. Just focus on the parts that you find challenging.
- Choose another problem and repeat the steps above. The more problems you work on, the more chunks you build in your mental library.
- Practice active repetition: review the problems mentally while you're doing something active (e.g. walking or exercising). This improves your ability to recall the solutions anytime, anywhere.



Once you've grasped the basics, you can improve by overlearning or interleaving.

- Overlearning means to keep learning/practicing something until it becomes automatic. This is useful for perfecting a piano piece or a tennis swing.

KEY QUOTES

"A little bit of work on something that feels painful can ultimately be very beneficial."

"Once you start working on a math or science problem, you'll notice that each step you complete signals the next step to you."

"A library of these chunks gives you an understanding of fundamental concepts in a way nothing else can."

- For math and science, it's more useful to use interleaving, i.e. do a mix of problems that involve different strategies or approaches.

- (i) Focus on 1 topic for some time, then shift to a totally different topic. Alternatively, open to a random page of the book and start working on a problem there.
- (ii) Prepare index cards with questions on one side and solutions on the other. Shuffle the cards and test yourself with random questions involving different approaches. Initially, do this at a table and write down your solution by hand before you review the answer. Once you're confident, you can review the cards anywhere (e.g. when you're out for a walk) to strengthen the chunking effect.



To understand and internalize complex or abstract math/science ideas, seek to simplify and personalize them.

- Formulas and equations are merely ways to abstract and simplify concepts. Like poetry, the numbers and symbols hold layers of deeper meaning. Try to understand the math/science equations so thoroughly that you can see them in your mind's eye. For instance, a physicist doesn't merely see the symbol for "mass"; he sees and senses the inertia of, say, a huge boulder.
- To grasp a complex concept, look for simple explanations or analogies. When you can break down a complex idea into its core elements, you'll have a much deeper understanding of it.
 - (i) Einstein used to imagine himself as a photon moving at the speed of light and how other photons might perceive him. Charles Darwin imagined himself explaining an idea in the simplest terms to a stranger.
 - (ii) Imagine yourself explaining the concept to a 10-year old—how could you express it in the simplest possible way?



Learning doesn't happen in a straight line. Sometimes, you must step back to step forward. After a period of learning, you may suddenly feel confused because your mind is synthesizing the material and restructuring your knowledge. You'll usually emerge from the confusion with a leap in understanding.

KEY QUOTES

"Testing is itself an extraordinarily powerful learning experience."

"There are hidden meanings in equations, just as there are in poetry."

"Simple explanations are possible for almost any concept, no matter how complex."

"The learning process is all about working your way out of confusion. Articulating your question is 80 percent of the battle. By the time you've figured out what's confusing, you're likely to have answered the question yourself!"

—Kenneth R. Leopold



Improving your Memory



When you first learn or notice something, it's stored in your working memory. To move it to your long-term memory, you must repeat the idea and make it memorable.

- Instead of rote memorization, use all 5 senses to create neural hooks to “hang” your ideas. For example, you'll remember a pig more clearly if you see, feel, hear and smell one (instead of merely seeing the word “pig”). Use the memory tricks and techniques below to tap into your brain's huge visual and spatial memory.



The “memory palace technique” is about anchoring key concepts visually to a familiar scene, such as your home or the journey to work.

- The key is to link an idea with a memorable image. For example, you can use a memory mnemonic to remember the sequence of minerals from the softest to the hardest: In front of the house, there's a Toad (talc) carrying a Giant (gypsum) which is holding a Can of Coke (calcite). You step through the Front door (fluorite) to see an Alligator (apatite) in the living room, playing the Organ (orthoclase). In the dining hall a Queen (quartz) is spinning a Top (topaz), eating Cheese (corundum) and sitting on a Deer (diamond).
- Once the images are vividly lodged in your mind, you can recall the sequence easily. Vivid, unusual and funny images leave the biggest impressions.



You can also use music, tunes and rhymes to remember formulas and equations. Search online for “quadratic formula” and “song” for existing samples, or make up your own!



Metaphors and analogies help to connect a new idea to an idea that we're already familiar with, making it easier to understand or recall. For example, you can explain electrical currents by comparing it to flowing water.



Use spaced repetition with all 5 senses to better encode ideas in your long-term memory. For instance, you can hand-write the symbol for density (ρ) on 1 side of an index card and write the key information (e.g. “measured in standard units of kilograms

KEY QUOTES

“Memory tricks allow people to expand their working memory with easy access to long term memory.”

“One of the best things you can do to not only remember but understand concepts in math and science is to create a metaphor or analogy for it—often, the more visual, the better.”

per cubic meter") at the back. As you do so, say it out loud and imagine a heavy 1m-by-1m luggage. Look at the symbol and recall what you know about p . Once you can do it without peeking at the back, put the card away. Revisit the card a few days later, then extend the period between each review.



Create meaningful groups to connect unrelated items:

- Acronyms: To remember the 4 plants to ward off vampires (GRHM—Garlic, Rose, Hawthorn, Mustard) think of a GRAHAM cracker.
- Associate numbers with memorable events, e.g. your birthday.
- String separate concepts into a memorable story or sentence. For example, the sentence "King Henry died while drinking chocolate milk" can help you to remember the decimal system (Kilo=1,000; Hecto=100; deca=10; while=1; deci=0.1; centi=0.01; milli=0.001).



Exercise regularly as it helps with neural growth and connections, thus improving your memory and learning abilities.



Overcoming Procrastination



Learning math/science is like laying a brick wall. You need brief periods of focused study to lay the neural bricks, with some time in between for the mortar to dry. Procrastination hinders learning since: (i) it leaves you with no time to explore or synthesize the concepts, and (ii) it's harder to absorb new material when you're stressed.



Procrastination is a keystone bad habit that affects all aspects of your life.

- We procrastinate to avoid anticipated discomfort. Paradoxically, the more you put off something, the more painful it becomes when you eventually do it. You also start to lose confidence in yourself over time.
- Habits involve 4 parts: A cue sparks a routine automatically, which gives you a reward that leads to certain beliefs about yourself or your habits. For instance, the sight

KEY QUOTES

"Procrastination is a single, monumentally important 'keystone' bad habit."

of your math homework (cue) may prompt you to start playing video game (routine). This offers temporary pleasure/relief (reward) and you start to think you can never learn to like math (belief), or create excuses (e.g. "If I study too early I'll forget what I learn").



- To change a habit, you must reprogram your response to the cue or avoid the cue altogether.
 - (i) Track your daily activities for 1-2 weeks. Each time you fail to finish what you intended, identify the procrastination routine and the cue(s) that trigger it.
 - (ii) Make a plan to break the routine and establish a new one, e.g. leave your phone in the car and do your math homework in the library where there're no distractions. Do 25 minutes of focused work using the Pomodoro technique (details below), then reward yourself immediately with something you enjoy.
 - (iii) The goal is to train your brain to anticipate a reward at the end of your new routine. Over time, you'll become better at what you do and start to enjoy it more.
 - (iv) Build supportive beliefs by hanging out with positive people with a "can-do" mindset.



The Pomodoro technique is one of the best procrastination fixes:

- Find a quiet environment, remove all distractions (e.g. turn off your phone, web browsers, and email prompts), and set a timer for 25 minutes. Work on 1 task with 100% focus until the time is up. Then, reward yourself immediately.
 - (i) Make sure you work on *1 task at a time*. Don't multitask as you'll be diluting your attention.
 - (ii) After each session, *reward yourself* with a break or something you enjoy, e.g. surf the web or play a video game.
- To get in flow, focus on the process, not the output. Don't worry about finishing a task. Just work with full

KEY QUOTES

"Procrastination can be like taking tiny amounts of poison. It may not seem harmful at the time. But the long-term effects can be very damaging."

"Focus on the process (the way you spend your time) instead of the product (what you want to accomplish)."

concentration for 25 minutes without stress or judgment, and you may be amazed by how much you can accomplish.

- Break down your long-term goals into bite-sized pieces so you can tackle them in short intense bursts, one day at a time.
 - (i) List down the key tasks you wish to accomplish each week.
 - (ii) Each evening, select a few key items you can reasonably tackle the next day. This gives your brain some time to figure out how to get things done.
 - (iii) Start each day with the most important or difficult task while you're still fresh.



You may be hyper-productive when you're stressed or in flow, but it's not sustainable. To avoid burnout, plan for both work and rest/play.

- Use a calendar to schedule time in advance for your daily tasks.
- Plan your stop time (e.g. 5pm each day). Then, make sure you stop work once you finish your list or reach the stop-time, so you have time to relax and rejuvenate.
- Include play time after your focus sessions and at the end of each day. This gives you something to look forward to, and helps to reduce procrastination.



Procrastination isn't always bad, and there are benefits to waiting.

- Math experts (professors and graduate students) take 45 seconds longer on average to think about a problem before solving it. Novices (undergraduates) tend to rush in after only 30 seconds of thinking. Take some time to pause and reflect before taking action. This gives your brain time to access its library, see the bigger picture and connect chunks of ideas/information.
- Prepare a list of problems you wish to solve in advance. If you know you're procrastinating, write down the equation(s) required for the easiest problem and carry it around with you. This allows your mind to work on the problem in diffuse mode while you're procrastinating, transforming it into "positive procrastination".

KEY QUOTES

"It's important to transform distant deadlines into daily ones."

"Staying in the zone too long will send you toward burnout."

"Planning your quitting time is as important as planning your working time."

"Planning your life for 'playtime' is one of the most important things you can do to prevent procrastination."

"Just as focused attention isn't always good, seemingly nasty habits of procrastination aren't always bad."



Use these additional tips to motivate yourself:

- Hold back all rewards and dish them out only when you've have finished a task/session.
- Place your list of tasks somewhere visible, and check off the items as you complete them. Celebrate every achievement.
- Surround yourself with pictures that remind you of your goals or desired future.
- Everyone has bad days when they feel negative about work or themselves. If you ignore those feelings and do the work anyway, the feelings will disappear on their own.



To get started, identify a task that you've been procrastinating on, break it down into smaller parts, and start tackling the first piece. Test out the tips/approaches above to find out what works for you.



Redesign your Brain and Build your Confidence



Your brain and mental capacity are not fixed. You can change your brain by changing how you think. The more you practice something, the more chunks and neural highways you build, and the easier it is to learn and solve problems.

- Each chunk of knowledge helps you to build new chunks, e.g. knowledge of living cells helps you to study biological evolution.
- You can also transfer chunks of knowledge, e.g. once you know how to learn math and science, you can use the knowledge to learn virtually anything.



What matters isn't how smart you are, but how you train your mind to learn in a deliberate and consistent way. Many science and math "geniuses" actually did badly in school and succeeded in spite of it.

- Learn at your own pace. Set aside enough time to study complex materials, and do lots of deliberate practice to build the right foundation.

KEY QUOTES

"Accomplishing a lot of difficult tasks is like eating a salami. You go slice by slice—bit by bit. Cheer every accomplishment, even the tiniest ones."

"Persistence is often more important than intelligence. Approaching material with a goal of learning it on your own gives you a unique path to mastery."

KEY QUOTES

- Take responsibility for your own learning. Don't rely on your teachers to instruct or motivate you. Ask questions to deepen your learning, do your own research and practice, and seek inputs/advice from others.
- Embrace failures and mistakes as a part of the learning process and a chance to learn something new.



Use these tips to prepare for your tests and exams.

- Prepare well in advance:
 - (i) Do your homework right: understand the text and solutions, discuss and check the solutions with your classmates, and try to solve every problem on your own.
 - (ii) Prepare well: seek help if required, participate actively in homework group discussions, and do all the tests in the study guides. Organize your study materials neatly so you can review them quickly just before a test/exam.
 - (iii) Use mini-tests (such as using recall and flash cards) to accelerate chunking.
- During a test:
 - (i) Use the "hard-start-jump-to-easy technique": Start with the hardest question first, but move on quickly after 1-2 minutes if you get stuck. Tackle the easier questions next so you can get as much covered as possible while your brain continues to work on the tough problems in the background.
 - (ii) Occasionally look away to shift your attention before returning to check your answers.
 - (iii) If you feel panicky, reframe the situation by telling yourself, "I'm excited to do my best!" Alternatively, breathe deeply and slowly to calm yourself down.
- Most importantly, get enough sleep before the test, or it will negate all your hard work. When you sleep, the brain flushes out toxins, consolidates knowledge/memories into chunks, works on tough problems and finds meaning in what you're learning. It also rehearses and reinforces what's important and prunes away what's trivial. A lack of sleep reduces your ability to think, focus, understand new concepts, and hurts your long-term health.

"I have not failed. I've just found 10,000 ways that won't work."
—Thomas Edison